

EVE FREEMAN

LEAD SOFTWARE ENGINEER

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PROFILE

Lead software engineer with 10+ years building cloud-native, data-intensive platforms. Led 11 engineers across 3 teams delivering roadmap features and raising engineering standards. Deep in databases (graph + relational), distributed systems, and secure, cost-aware cloud architecture. Pragmatic about AI adoption—governance, observability, and spend controls built in.

PROFESSIONAL EXPERIENCE

Lead Software Engineer, Fannie Mae
Reston, VA

Apr 2016 — Present

- **Engineering Leadership:** Led 11 engineers across 3 teams; partnered with product to define and deliver roadmap; set and enforced engineering standards.
- **Mentorship & Quality:** Mentored engineers across levels; drove design/code reviews and incident retros; improved quality and delivery predictability.
- **AI Gateway / Control Tower:** Built an enterprise AI request-management platform across multiple AI providers with observability, cost attribution, and governance controls (kill switches, rate limits, automated cost alerts) to keep usage safe and on budget.
- **Agentic Engineering:** Designed and built autonomous AI agent workflows to automate complex, multi-step engineering and operational tasks; integrated tool-use patterns and retrieval-augmented context to accelerate delivery and reduce manual work.
- **Data Lineage Platform:** Migrated the lineage platform from Oracle to Neo4j; improved complex graph-traversal query performance 10x+ by optimizing Cypher and the graph model for enterprise-scale relationships; reduced infrastructure costs and improved developer experience.
- **Custom Graph Library:** Built a Java in-memory graph library (Roaring Bitmaps) to offload traversal-heavy logic from Neo4j stored procedures and remove performance bottlenecks.

Open Source Contributor / Training / Meetup Organizer, Neo4j (freelance consultant)
Virginia

Apr 2013 — Oct 2019

- **Cypher Language Contributions :** Contributed to Neo4j's Cypher query language during the 2.0 era, implementing several math aggregation functions and the **reduce** function that are still parts of the language. These contributions are used by developers worldwide in production graph database queries.
- **Course Development :** Developed comprehensive curriculum for advanced Cypher course, creating structured learning materials for complex graph query patterns
- **Multi-Level Instruction :** Taught Neo4j courses covering modeling, Cypher fundamentals, and advanced Cypher techniques to diverse audiences
- **International Training :** Conducted training sessions including one fully in Spanish in Mexico City, which I translated and delivered, demonstrating ability to adapt technical content for different languages and cultures
- **Long-Term Partnership :** Maintained freelance training relationship with Neo4j for several years, building expertise in graph database education and knowledge transfer

CTO of Emerging Technology, Information Analysis Inc. (now Tellenger)

Aug 2009 — Apr 2016

Fairfax, VA

- **Government Technology Leadership** : Led development of web and mobile applications for NASA and DHS, working with complex requirements and security constraints
- **Graph Database Expertise** : Early adopter of graph databases for relationship mapping and knowledge management use cases
- **Natural Language Processing** : Applied NLP techniques to extract insights from unstructured text data
- **Document Services** : Integrated Adobe Document Services for enterprise document processing and management

Software Engineer, Metalsa

Jul 2004 — Mar 2009

Roanoke, VA

- **ERP Implementation** : Contributed to Oracle ERP system implementation for manufacturing operations, integrating business processes across the organization
- **Manufacturing Automation** : Developed software solutions to automate manufacturing workflows and improve operational efficiency
- **Business Intelligence** : Built reporting systems to provide insights into manufacturing operations and business metrics

EDUCATION

Western Governors University

Jul 2025

Bachelor of Science, Cybersecurity & Information Assurance, Salt Lake City, UT

Thomas Jefferson High School for Science and Technology

Jun 2000

Computer Systems, Annandale, VA

SKILLS

Languages: Java, Python, Go, TypeScript, Rust

Frontend: Angular, React, Svelte

Backend: Spring Boot, FastAPI, Go (minimal deps; net/http)

Datastores: PostgreSQL, DynamoDB, Redis, Neo4j, MongoDB, Elasticsearch, S3

CI/CD: Maven, npm, GitHub, GitLab, SonarQube

Infrastructure: AWS, Terraform, Docker, Kubernetes

CERTIFICATIONS

AWS Certified Solutions Architect – Professional

Jan 2024

AWS Certified DevOps Engineer – Professional

Feb 2024

AWS Certified Machine Learning – Specialty

Dec 2023

AWS Certified Data Analytics – Specialty

Oct 2023

AWS Certified Database – Specialty

Sept 2023

Linux Foundation Certified Sysadmin (LFCS)

Nov 2024

Linux Foundation Certified FinOps Engineer

Oct 2024

CompTIA CySA+

Apr 2025

CompTIA Security+

Feb 2025

ADDITIONAL INFORMATION

Top 1% answerer in Go, Neo4j, and MongoDB on StackOverflow: <https://stackoverflow.com/users/683201/eve-freeman>

Full certifications list on Credly: <https://www.credly.com/users/eve-freeman/>

GitHub: <https://github.com/freeeve/>

Websites: <https://evefreeman.com>, <https://deeplibby.com>

Hobbies: Reading (especially litrpg, sci-fi, fantasy, historical fiction, and westerns), collecting library cards, chess.

Languages: English (native), Spanish (fluent), Japanese (duolingo/skritter-dabbler)